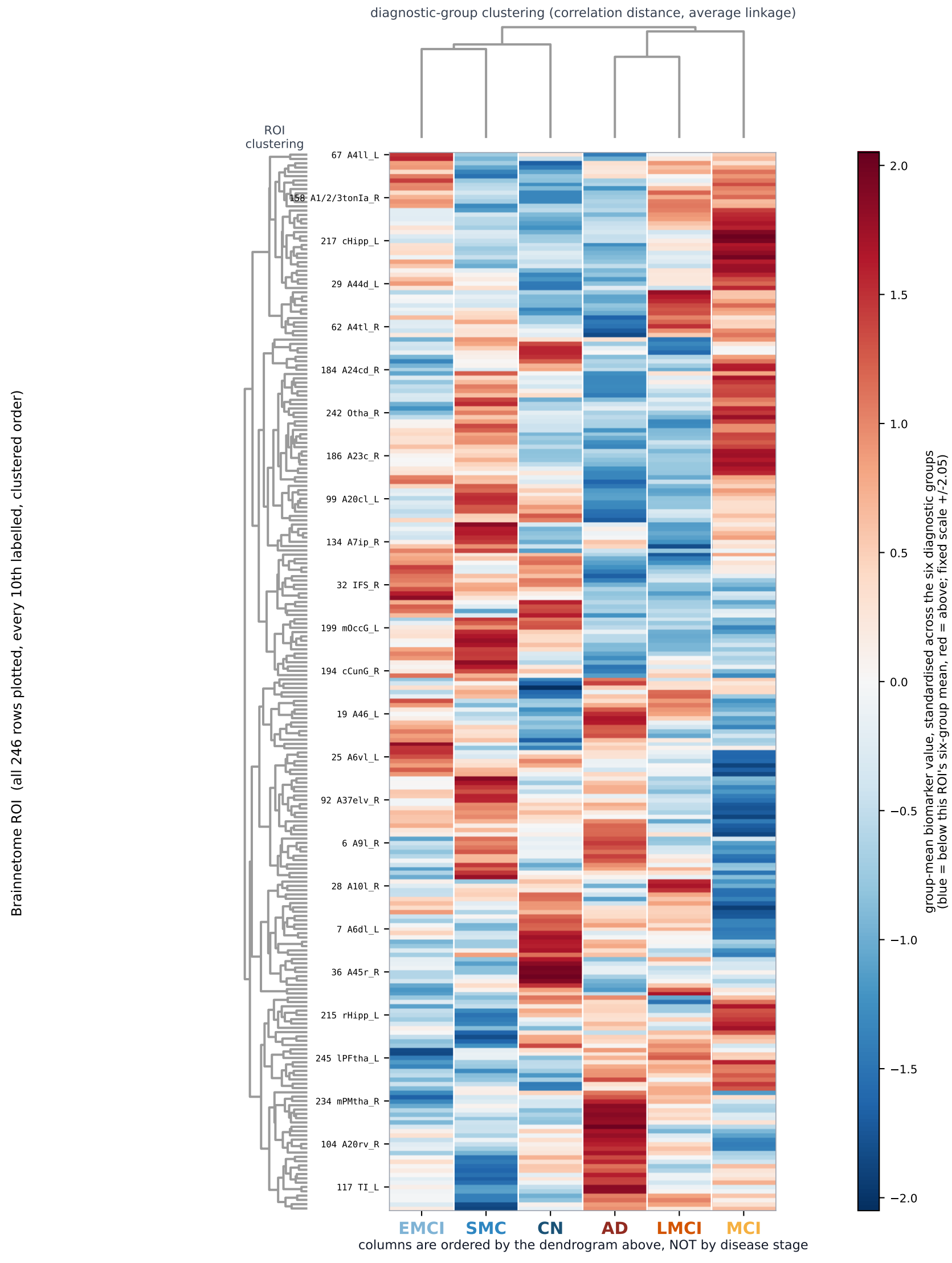
Hierarchical Clustering of Alzheimer's Diagnostic Groups
Regional amplitude of low-frequency fluctuation (mALFF)

cells are the MEAN mALFF of the subjects in each diagnostic group (n = 20/20/20/16/20/20), standardised across the six groups per ROI (ddof = 1)
groups clustered by correlation distance + average linkage over the 246-D profile; ROIs by euclidean distance + average linkage

group P-score = -log10(q) from the MULTIVARIATE stage-vs-rest permutation test (24-test family)

max ROI P = largest ROI-level P-score for that group -- DESCRIPTIVE, not a test | both are far below the 1.301 significance threshold

	EMCI	SMC	CN	AD	LMCI	MCI
group P-score	**0.012**	**0.403**	**0.154**	**0.738**	**0.119**	**0.738**
max ROI P	0.259	0.284	0.284	0.652	0.284	0.284



clinical progression order: CN → SMC → EMCI → MCI → LMCI → AD

bootstrap stability of the branches (1000 resamples of subjects within groups): {LMCI,AD} 46%; {CN,SMC,EMCI} 25%; {SMC,EMCI} 25%

Heatmap shows BIOMARKER ACTIVITY; the top row shows STATISTICAL SIGNIFICANCE. They are different quantities. The tree clusters on biomarker activity.

Descriptive only. Branch heights are distances, not p-values. No group reaches q < 0.05 and no ROI x stage test in this project does either.